

YASH KUMAR

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Skills

Programming Languages: Python, Java, C/C++, JavaScript • **Machine Learning & AI:** Scikit-learn, TensorFlow, PyTorch, Keras, Computer Vision: OpenCV, CNN • **Natural Language Processing:** NLTK, spaCy, Transformers (BERT, GPT) • **Cloud Technologies:** AWS, GCP, Azure • **Databases:** MySQL, PostgreSQL, MongoDB • **Version Control:** Git, GitHub, Bitbucket • **Development Tools:** Docker, Kubernetes, Jenkins **LLM tooling & methods:** LoRA, quantization (bitsandbytes), Hugging Face Transformers, retrieval-augmented generation (RAG), LangChain, prompt engineering, evaluation/benchmarks

Experience

Alignerr — Machine Learning Engineer (Freelance) | Remote | **01/2025 – 05/2025**

- Led annotation strategy and trained labeling teams using Labelbox/CVAT; established QA workflows, inter-annotator agreement metrics (e.g., Cohen's kappa), and escalatio.
- Led annotation strategy and trained labeling teams, implementing QA and inter-annotator agreement checks to improve label quality and reduce noisy labels.
- Built automated validation and data-curation frameworks and test harnesses to generate and verify edge cases, ensuring solver reliability across diverse data distributions. Automated image-processing pipelines (OpenCV, PyTorch/TensorFlow), deployed inference APIs (FastAPI/Flask) with CI/CD, and wrote optimized SQL ETL for monitoring and retraining.

Outlier AI – Machine Learning Expert Remote **08/2024 - 12/2024**

- Led annotation program and QA: authored schemas and guidelines, trained labelers, implemented inter-annotator agreement and review workflows to reduce noisy labels and improve dataset consistency.
- Partnered with ML engineers to integrate curated datasets into training workflows and measure impact on model performance.
- Designed and optimized complex SQL queries for efficient data extraction and transformation supporting ML reporting ion quality.

Education

Bachelor of Science in Information Technology

Lovely Professional University, Punjab

June 2021 – July 2024

Projects

• **SmartShop** — Lead ML Engineer (Project Lead) | **01/2023 – 05/2023**

- Benchmarked collaborative (SVD/NMF), embedding DNNs, hybrid content-collaborative, and sequence models (LSTM/GRU); engineered features from 100K+ product transactions.
- Deployed FastAPI real-time serving with Redis and CV-based visual search; achieved 78% test accuracy, +35% engagement, 10K+ DAU — [demo](#)

Voice Recording Analysis — Lead Engineer | **09/2024 – 12/2024**

- Led end-to-end development and production deployment of a multimodal voice-analysis pipeline (ASR, diarization, emotion detection, summarization), achieving 95% transcription accuracy and real-time multilingual processing. [Code](#)

PUBLICATIONS

Y. Kumar, H. Ahmed, H. K. Sinha — “**SmartShop: Personalized Grocery Item Recommendation System**”, **ICONIC2023**.

Y. Kumar, H. Ahmed, V. Subhash — “**Prediction Model for Flight Fare: A Generic Review**”, **ETEMPS 2023**